

Geodesics and hyperbolicity for the half-plane models of taxicab hyperbolic space

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Abstract

We develop a new geometric space by blending The half-plane model for the hypergolic plane with taxicab geometry. Starting with a taxicab norm on each tangent space, the lengths of curves are analyzed, allowing us to determine geodesics, and from these, a distance function. Then we show that these spaces are hyperbolic in the sense of Gromov.

Background and Introduction

Given $m \in [0, 1]$, our space is $(\mathbb{H}_m^2, \|\cdot\|_m)$ where

$$\mathbb{H}_m^2 = \{x = (x_1, x_2) \in \mathbb{R}^2 : x_2 > mx_1\} \text{ and } \|v\|_{m,x} = \frac{\|v\|_T}{dT(x, \ell_\infty)} = \frac{|v_1| + |v_2|}{x_2 - mx_1}.$$

Given a curve $\gamma : [0, 1] \rightarrow \mathbb{H}_m^2$, its length can be calculated:

$$\mathcal{L}(\gamma) = \int_0^1 \|\gamma'(t)\|_m dt = \int_0^1 \frac{|\gamma_1'(t)| + |\gamma_2'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt,$$

and using this, among the absolutely continuous curves from p to q , minimizers can be found, thus determining the geodesics and a distance function for this space. (See middle column.)

With a distance function in hand, Gromov hyperbolicity can be analyzed. Given three points x, y , and z , the Gromov product is

$$G(x, y, z) = \frac{1}{2}(d(x, z) + d(y, z)) - d(x, y).$$

The Gromov hyperbolicity inequality says that a space is δ -hyperbolic if there exists a δ such that for all points x, y, z, w ,

$$G(x, y, w) \geq \min\{G(x, z, w), G(y, z, w)\} - \delta$$

Defining the function

$$f(x, y, z, w) = \min\{G(x, z, w), G(y, z, w)\} - G(x, y, w),$$

if f is bounded by δ then the space is δ -hyperbolic. We show that $\mathbb{H}_m^2$ is δ -hyperbolic. (See the rightmost column.)

Future Work

1. Resolve $m = 1$ case
2. Consider higher dimensions
 - horizontal hyperplane at infinity
 - slanted hyperplane at infinity
3. quotient spaces

References

1. Fish, A. and Helliwell, D. (2025) *A Poincaré ball model for taxicab hyperbolic geometry*, Involve.

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Geodesics

Comparing the Length of a Curve λ to the Length of a General Curve γ

Here we're going to compare the lengths of two curves λ and γ . λ is the concatenation of two line segments $\sigma_{p,a}$ and $\sigma_{a,q}$, where $\sigma_{a,q}$ is a coordinate-horizontal line segment from point p to point a , and $\sigma_{p,a}$ is a coordinate-vertical line segment from point a to point q . Point a has coordinates (p_1, q_2) . γ is any doubly monotonic curve who's second coordinate γ_2 is less than or equal to q_2 at every point.

By plugging the coordinates of a into the length formulas for coordinate-horizontal and coordinate-vertical lines, we get out the lengths of $\sigma_{p,a}$ and $\sigma_{a,q}$:

$$\sigma_{p,a} = \ln \left| \frac{q_2 - mp_1}{p_2 - mp_1} \right|$$

$$\sigma_{a,q} = \frac{1}{m} \cdot \ln \left| \frac{q_2 - mp_1}{q_2 - mq_1} \right|.$$

Now, we'll compute the length of γ :

$$\begin{aligned} \mathcal{L}(\gamma) &= \int_0^1 \frac{|\gamma_1'(t)| + |\gamma_2'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\ &= \int_0^1 \frac{|\gamma_1'(t)| + m|\gamma_1'(t)| + |\gamma_2'(t)| - m|\gamma_1'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\ &= \int_0^1 \frac{|\gamma_1'(t)| + m|\gamma_1'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt + \int_0^1 \frac{|\gamma_2'(t)| - m|\gamma_1'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \end{aligned}$$

First, we'll look at the coordinate-horizontal component of γ :

$$\begin{aligned} &\int_0^1 \frac{|\gamma_1'(t)| + m|\gamma_1'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\ &= (1+m) \cdot \int_0^1 \frac{|\gamma_1'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\ &= (1+m) \cdot \int_0^1 \frac{\gamma_1'(t)}{\gamma_2(t) - m\gamma_1(t)} dt \\ &\geq (1+m) \cdot \int_0^1 \frac{\gamma_1'(t)}{q_2 - m\gamma_1(t)} dt \\ &= \frac{1+m}{-m} \cdot \int_{t=0}^{t=1} \frac{1}{u} du \\ &= \frac{1+m}{-m} \cdot \ln |q_2 - m\gamma_1(t)|_0^1 \\ &= \frac{1+m}{-m} \cdot \ln \left| \frac{q_2 - mq_1}{q_2 - mp_1} \right| \\ &= \frac{1+m}{m} \cdot \ln \left| \frac{q_2 - mp_1}{q_2 - mq_1} \right| \end{aligned}$$

We used a u -substitution, where $u = q_2 - m\gamma_1(t)$. This length is greater than the length of $\sigma_{a,q}$.

Next, we'll look at the coordinate-vertical component of γ :

$$\begin{aligned} &\int_0^1 \frac{|\gamma_2'(t)| - m|\gamma_1'(t)|}{\gamma_2(t) - m\gamma_1(t)} dt \\ &= \int_0^1 \frac{\gamma_2'(t) - m\gamma_1'(t)}{\gamma_2(t) - m\gamma_1(t)} dt \\ &= \int_0^1 \frac{1}{u} du \\ &= \ln |\gamma_2(t) - m\gamma_1(t)|_0^1 \\ &= \ln |q_2 - mq_1| - \ln |p_2 - mp_1| \end{aligned}$$

So, the length of λ is less than the length of γ .

General length formula for $\mathbb{H}_m$ is

$$|\ln(\frac{\mu - mp_1}{p_2 - mp_1})| + |\ln(\frac{\mu - mq_1}{q_2 - mq_1})| + \frac{1}{m} |\ln(\frac{\mu - mp_1}{\mu - mq_1})|$$

Hyperbolicity

We start analyzing hyperbolicity in $\mathbb{H}_m^2$ by using our distance equation with points x, y, z . The Gromov Product is expressed as so, where $\mu = \max\{p_2, q_2, \frac{|p_1 - q_1| + m(p_1 + q_1)}{2}\}$

$$\begin{aligned} G(x, y, z) &= \frac{1}{2} \left[\left(\ln \left(\frac{\mu - mx_1}{x_2 - mx_1} \right) + \ln \left(\frac{\mu - mz_1}{z_2 - mz_1} \right) + \frac{1}{m} \ln \left(\frac{\mu - mx_1}{\mu - mz_1} \right) \right) \right. \\ &\quad \left. + \left(\ln \left(\frac{\mu - my_1}{y_2 - my_1} \right) + \ln \left(\frac{\mu - mz_1}{z_2 - mz_1} \right) + \frac{1}{m} \ln \left(\frac{\mu - my_1}{\mu - mz_1} \right) \right) \right. \\ &\quad \left. - \left(\ln \left(\frac{\mu - mx_1}{x_2 - mx_1} \right) + \ln \left(\frac{\mu - my_1}{y_2 - my_1} \right) + \frac{1}{m} \ln \left(\frac{\mu - mx_1}{\mu - my_1} \right) \right) \right] \end{aligned}$$

We use the aforementioned definition of the Gromov Product in our hyperbolicity equation.

$$f(x, y, z, w) = \min\{G(x, z, w), G(y, z, w)\} - G(x, y, w),$$

Through much analysis, and rearranging of the μ terms, we become closer to finding an upper bound for the hyperbolicity function. The Ω terms in the brackets require close attention and a clever geometric interpretation to simplify. In each case, Ω is of the form $\mu(a, b) - ma_1$. We found Ω to be symmetric, which means a and b can be interchanged.

$$f(x, y, z, w) \leq \frac{1}{2} \ln \left[\frac{\Omega(x, y)\Omega(z, w)}{\max\{\Omega(x, z)\Omega(y, w), \Omega(x, w)\Omega(y, z)\}} \right] + \frac{3}{2m} \ln \left(\frac{1+m}{1-m} \right)$$

The geometric interpretation

Using two sets of assumptions about the values of μ inside Ω terms, we come to a conclusion involving estimates about vertical distances inside the numerator and denominator. This information can be used to find a case in which the fraction evaluates to value with just constant terms, therefore giving a bound for the function f and showing that space is hyperbolic.

We first start with expanding the Ω terms and noting certain symmetrical properties. Expanded fully, we see the Ω terms are some combination of x, y, z , or w , in the form of $\mu(p, q) - mp_1$. Assuming $\mu = p_2$, the term $p_2 - mp_1$ is a measure of the vertical distance from p to the slanted line at infinity. So let us make an assumption about the numerator in the fraction to make these terms more understandable. Let $\mu(x, z) = x_2$, and $\mu(z, w) = z_2$. Therefore $x_2 \geq y_2$, and $x_2 \geq \frac{|x_1 - y_1| + m(x_1 + y_1)}{2}$. From these inequalities we can see that y must be below x , and y has a horizontal limit at some point left or right from x . Let us rearrange the inequality so we are in terms of Ω , where $\Omega = \mu(x, y) - my_1$. We use cases to remove the absolute value bars, and start Case 1 by assuming $y_1 \leq x_1$.

$$\begin{aligned} &y_1 \leq x_1 \\ &x_2 \geq \frac{x_1 - y_1 + m(x_1 + y_1)}{2} \\ &2x_2 \geq x_1(1+m) + y_1(m-1) \\ &y_1(1-m) \geq -2x_2 + x_1(1+m) \\ &y_1 \geq -x_2 \frac{-2}{1-m} + x_1 \frac{1+m}{1-m} \\ &-my_1 \leq x_2 \frac{-2}{1-m} - x_1 \frac{1+m}{1-m} \\ &x_2 - my_1 \leq x_2 \left(\frac{2m}{1-m} \right) - mx_1 \frac{1+m}{1-m} + x_2 \\ &x_2 - my_1 \leq x_2 \left(1 + \frac{2m}{1-m} \right) - mx_1 \frac{1+m}{1-m} \\ &x_2 - my_1 \leq x_2 \left(\frac{1+m}{1-m} \right) - x_1 \left(\frac{1+m}{1-m} \right) \\ &x_2 - my_1 \leq (x_2 - mx_1) \left(\frac{1+m}{1-m} \right) \end{aligned}$$

(LUKA's NOTE - go through the p and q case to fully display the symmetry in the argument to wrap it up.) Therefore with the assumptions about μ made, we have this geometric interpretation; the vertical distance of y to the line at infinity is smaller or equal to the vertical distance of x , and is no farther away horizontally due to the limiting term $\frac{1+m}{1-m}$. Now we evaluate Case 2 in the absolute value to see if there is a limiting factor on y_1 when it is greater than x_1 . First let $y_1 \geq x_1$. Note that $x_2 - my_1 \leq x_2 - mx_1$, since we assumed $y_1 \geq x_1$ which is certainly less than $(x_2 - mx_1)(\frac{1+m}{1-m})$, as we saw earlier. In a final note, we observe some symmetry among terms in Ω (MORE ON THIS!) For the term $\Omega(x, y)$ in the numerator, we have the term $\Omega(x, w)$ in the denominator that bounds the upper term by $\frac{1+m}{1-m}$. (due to symmetry) we apply this bound on all other terms finding that the entire expression simplifies to:

$$\left(\frac{1+m}{1-m} \right)^4$$

Plugging this back in with the rewritten Gromov Hyperbolicity equation shows that the function f is bounded by